

PERMUTATIONS WITH NO WHOLE-NUMBER AVERAGES

VAMSHI JANDHYALA

ABSTRACT. Call a permutation of $1, \dots, n$ good if no block of consecutive entries, other than the whole permutation, of length at least two has an integer average. A question on MathOverflow asks whether, for odd n , good permutations exist exactly when n is a Mersenne prime. An answer there shows that n must have the form $2^m - 1$. We prove that the asker's example $1, n-1, n, n-3, n-2, \dots, 2, 3$ is good exactly when $n = 2^m - 1$ is prime, and an exhaustive search shows that the composite cases $n = 15$ and $n = 63$ have no good permutation at all. The theorems, including the answer's, are formally verified in Lean 4.

1. THE QUESTION

Take the numbers $1, \dots, 7$ in the order

$$1, 6, 7, 4, 5, 2, 3.$$

Pick any run of consecutive entries, of length between 2 and 6, and average it: $6, 7$ gives $6\frac{1}{2}$, $1, 6, 7$ gives $4\frac{2}{3}$, $7, 4, 5, 2$ gives $4\frac{1}{2}$. None of the averages is a whole number. The whole permutation is the only exception, and it has to be: its average is $(1 + \dots + 7)/7 = 4$. Before reading on, try to do the same with $1, \dots, 5$; it cannot be done.

Philip Weiss asked on MathOverflow in August 2026 [1] about such arrangements. Call a permutation $a_1, \dots, a_n$ of $1, \dots, n$ *good* if every block $a_s, a_{s+1}, \dots, a_e$ with $2 \leq e - s + 1 < n$ has a non-integer average. For even n good permutations are easy, as Weiss noted in a comment: in $2, 1, 4, 3, \dots, n, n-1$ every block of even length has a half-integer average, and every block of odd length $L \geq 3$ has sum $\equiv \pm 1 \pmod{L}$. For odd n he observed that the permutation

$$(1) \quad 1, n-1, n, n-3, n-2, \dots, 2, 3$$

is good when n is a Mersenne prime $2^m - 1$, found by search that good permutations exist for odd $n \leq 41$ only when $n = 3, 7, 31$, and asked whether, for odd n , a good permutation exists if and only if n is a Mersenne prime.

Half of the question is settled. Building on a comment by te4, Sergiu Alexandru Bîsceanu proved in an answer [2] that n must be a Mersenne number.

Theorem 1 (Bîsceanu). *If $n \geq 3$ is odd and a good permutation of $1, \dots, n$ exists, then $n = 2^m - 1$ for some m .*

The other half, that a composite $2^m - 1$ such as $15 = 3 \cdot 5$ has no good permutation, is open. This note adds three things. Theorem 2 explains why the example (1) needs n prime: its only blocks with integer average are its prefixes whose length divides n . The structure in Bîsceanu's proof cuts the search space by half, which takes the exhaustive search to $n = 63$, the second composite Mersenne number (Theorem 3). And every theorem here, Theorem 1 included, is checked in Lean.

Theorem 2. *Let $n = 2^m - 1$ with $m \geq 2$. A block of length $L \geq 2$ in (1) has an integer average if and only if it is a prefix and L divides n . In particular (1) is good if and only if n is prime.*

Theorem 3. *For $n = 7$ and $n = 31$ there are exactly four good permutations: (1), its reverse, its complement $a_t \mapsto n + 1 - a_t$ and the reverse of the complement. For $n = 15$ and $n = 63$ there are none.*

So the conjecture holds for every odd $n \leq 63$, and for $n \leq 31$ the good permutation is unique up to these symmetries.

2. WHY ONLY $2^m - 1$

The proof of Theorem 1 looks at blocks whose length is a power of two. Fix a good permutation of odd length $n \geq 3$, and write $S(t, L) = a_t + \dots + a_{t+L-1}$ for the sum of the block of length L starting at position t .

Lemma 4. *If $2^k < n$ with $k \geq 1$, then every block of length 2^k sums to an odd multiple of 2^{k-1} . Consequently $a_{t+2^k} \equiv a_t \pmod{2^k}$ whenever both positions exist.*

Proof. For $k = 1$ the claim says that two neighbours have an odd sum, which is exactly the statement that their average is not an integer. Suppose the claim holds for k and $2^{k+1} < n$. A block of length 2^{k+1} is two blocks of length 2^k , so its sum is $2^{k-1}(o_1 + o_2)$ with o_1, o_2 odd. That is a multiple of 2^k . It is not a multiple of 2^{k+1} , because the block is proper and its average is not an integer, so it is an odd multiple of 2^k .

For the congruence, slide the block one place: $S(t+1, 2^k) - S(t, 2^k) = a_{t+2^k} - a_t$. Both sums are odd multiples of 2^{k-1} , so their difference is a multiple of 2^k . $\square$

For $n = 7$ the lemma says that neighbours have opposite parity, and that $a_{t+4} \equiv a_t \pmod{4}$. In 1, 6, 7, 4, 5, 2, 3 the pairs $(a_1, a_5), (a_2, a_6), (a_3, a_7)$ are (1, 5), (6, 2), (7, 3), each differing by 4.

Proof of Theorem 1. Let q be the largest power of two below n and write $n = q + s$. Since n is odd and $q \geq 2$ is even, s is odd and $1 \leq s < q$. For $1 \leq i \leq s$ the entries a_i and a_{i+q} are distinct numbers in $1, \dots, n$ that agree modulo q , by Lemma 4. Their difference is a nonzero multiple of q and less than $2q$ in size, so it is $\pm q$: the pair is $\{x_i, x_i + q\}$ for some $x_i \leq s$. Distinct i give distinct x_i , since each value occurs once. So the s pairs use up exactly the values $1, \dots, s$ and $q+1, \dots, n$, and the middle block $a_{s+1}, \dots, a_q$ is a permutation of $s+1, \dots, q$. Its average is $(s+q+1)/2 = (n+1)/2$, an integer. The middle block is proper, so its length $q-s$ must be 1, and $n = 2q - 1$. $\square$

When $n = 2^m - 1$ the middle block is the single entry at position $q = 2^{m-1}$, and the proof gives more than the shape of n .

Corollary 5. *If $n = 2^m - 1$ and a is good, then $a_q = q$ for $q = 2^{m-1}$, and $\{a_i, a_{i+q}\} = \{x, x+q\}$ for some x , for every $i < q$.*

Figure 1 shows this for $n = 31$. A good permutation is determined by its first $q - 1$ entries, since each later entry is the partner of the entry q places earlier. That halving is what makes the search in Section 4 possible.

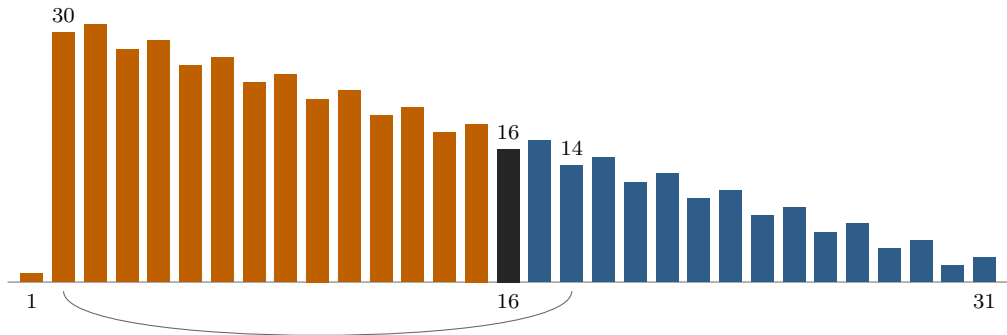


Figure 1. The permutation (1) for $n = 31$, as bars of height a_t . The middle entry is $a_{16} = 16$. Positions i and $i + 16$ hold two values 16 apart, as Corollary 5 requires of every good permutation; one such pair, $a_2 = 30$ and $a_{18} = 14$, is linked.

3. THE ASKER'S PERMUTATION

Weiss's example (1) has a closed form. Its first entry is $a_1 = 1$, and for $t \geq 2$

$$(2) \quad a_t = n + 2 - t - (-1)^t.$$

Read the right-hand side in two parts. The line $n + 2 - t$ falls from n at $t = 2$ to 2 at $t = n$; the term $-(-1)^t$ lowers each even position by one and raises each odd position by one, which swaps neighbours in pairs. For $n = 7$ the line gives 7, 6, 5, 4, 3, 2 at $t = 2, \dots, 7$, and the swaps turn it into 6, 7, 4, 5, 2, 3.

Proof of Theorem 2. Blocks away from the start. Take a block from position $s \geq 2$ to e , of length $L = e - s + 1 \geq 2$. By (2) its average is

$$n + 2 - \frac{s+e}{2} - \frac{\varepsilon}{L}, \quad \varepsilon = \sum_{t=s}^e (-1)^t \in \{-1, 0, 1\}.$$

If L is even, the signs cancel, $\varepsilon = 0$, and $(s+e)/2$ is a half-integer. If L is odd, $(s+e)/2$ is an integer and $\varepsilon = \pm 1$, so the average is an integer plus $\pm 1/L$ with $L \geq 3$. In neither case is it an integer.

Prefixes of odd length. Summing (2), the prefix of length $L = 2r + 1$ has sum $(2r + 1)(n + 1 - r) - n$. Its average is an integer exactly when L divides n .

Prefixes of even length. The prefix of length $L = 2v$ has sum $(2v - 1)(n + 1 - v) = (2v - 1)(2^m - v)$. Write $v = 2^j u$ with u odd. Since $v \leq n < 2^m$, we have $j < m$, so $2^m - v = 2^j(2^{m-j} - u)$ with $2^{m-j} - u$ odd. The sum is therefore 2^j times an odd number, while $L = 2^{j+1}u$. So L does not divide the sum, and no even prefix has an integer average.

Hence the blocks with integer average are the prefixes whose length $L \geq 2$ divides n . The prefix of length n is the whole permutation and does not count, so (1) is good exactly when n has no divisor L with $2 \leq L < n$, that is, when n is prime. $\square$

For $n = 15$ the two failures are $1 + 14 + 15 = 30 = 3 \cdot 10$ and $1 + 14 + 15 + 12 + 13 = 55 = 5 \cdot 11$, the prefixes of lengths 3 and 5. Figure 2 shows every block for $n = 15$ and $n = 31$.

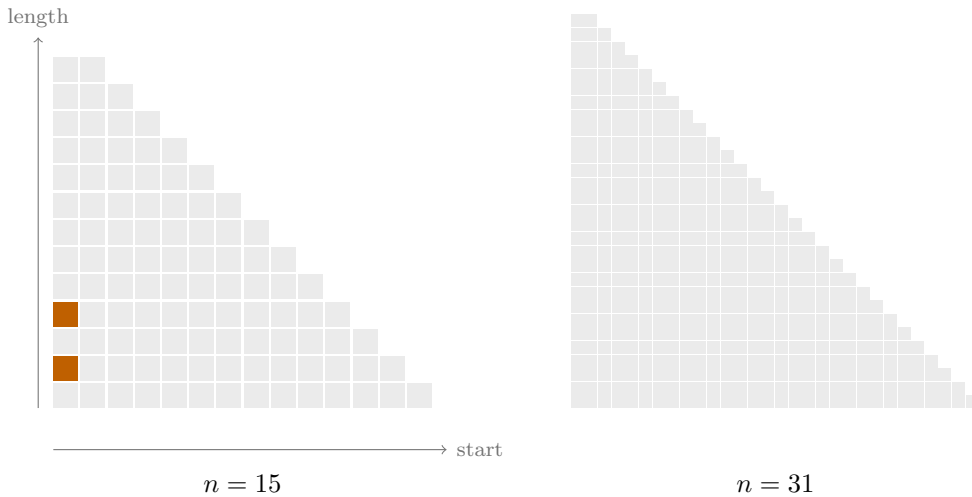


Figure 2. Every proper block of (1) of length at least 2: the cell in column s and row L is the block of length L starting at position s . Filled cells have an integer average. For $n = 15$ they are the prefixes of lengths 3 and 5, the proper divisors of 15; for the prime $n = 31$ there are none.

4. SEARCHING THE COMPOSITE CASES

For $n = 2^m - 1$, Corollary 5 reduces the search to choosing $a_1, \dots, a_{q-1}$. Choosing a_t also fixes a_{t+q} , so when a_t is placed one can test every block inside positions $1, \dots, t$ and every block inside positions $q+1, \dots, q+t$ at once. A depth-first search with these tests, followed by a full check of each completed permutation, gives the counts in Table 1.

n	good permutations	search nodes	plain search
7	4	23	4
15	0	855	0
31	4	85,387	4
63	0	106,605,579	—

Table 1. Exhaustive counts for $n = 2^m - 1$ with the reduction of Corollary 5, and with a plain search that uses no structure.

A second program, a plain depth-first search with no structural assumption, confirms the counts for $n = 7, 15, 31$ and finds no good permutation for any other odd $n \leq 41$, which reproduces Weiss’s search. It cannot reach $n = 63$. Its running time grows by a factor of about 2.5 with each step of n by two, so $n = 63$ would take about a month.

Remark 6. For $n = 15$ the obstruction lies in the block lengths 3 and 5 that divide n , as it does for (1). If the conditions for those two lengths are dropped and every other length is kept, the reduced search finds 320 permutations. A proof for composite $2^m - 1$ would have to use the divisors of n , which the powers of two in Lemma 4 do not see.

The next composite Mersenne number after 63 is 255, far beyond this search; the next prime case, $n = 127$, would take hours and would test whether the good permutation stays unique up to symmetry.

5. VERIFICATION

Theorem 1 (with Lemma 4), Corollary 5 and Theorem 2, including the fact that (1) is a permutation, are proved in Lean 4 with Mathlib [3], using only the standard axioms; ten planted errors in the statements and definitions all fail to compile. The searches of Theorem 3 are two C programs driven by a Python script, which also checks Theorem 2 directly for $n \leq 2047$; the reduced search runs only on the structure that Lean proves, and a planted error in it (skipping the lengths that divide n) is caught. Code and Lean files are at [4].

ACKNOWLEDGMENTS

The question and the permutation (1) are Philip Weiss’s; the 2-adic congruence is from a comment by te4, and Theorem 1 is Sergiu Alexandru Bîsceanu’s. Peter Taylor reported partial searches in a comment. The proof of Theorem 2, the searches and the Lean formalisation were carried out with Claude Opus 5.5 (Anthropic). The author is responsible for the content.

REFERENCES

- [1] P. Weiss, *Do only Mersenne primes have “good” permutations?*, MathOverflow question 514690 (2026), <https://mathoverflow.net/q/514690>.
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LONDON, UNITED KINGDOM

Email address: vamshi.jandhyala@gmail.com